Talk it through: friction

Language goal: I can **compare** static and kinetic friction.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · PUSH 3 N ON A CRATE THAT DOES NOT MOVE

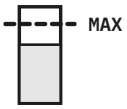


A ASKS "How big is the friction force?"

B SAYS "The crate does not move, so friction is ____ N. It matches the ____."

A CHECKS "So you're saying static friction can ____. I agree / disagree because ____."

ROUND 2 · SWITCH · THE LIMIT IS 6 N



B ASKS "What happens when the push goes past 6 N?"

A SAYS "The crate starts to ____. Then the friction is ____ friction."

B ADDS "I want to add that kinetic friction is usually ____ than the limit."

ROUND 3 · BOTH · 10 N PULL, 4 N KINETIC FRICTION, 2 kg



A "The net force is ____ − ____ = ____ N."

B "So the acceleration is ____ ÷ ____ = ____ m/s²."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Static friction is different from kinetic friction because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

FRICTION 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: friction

17 C

DIRECTIONS · Static limit 8 N, kinetic 5 N unless stated. Copy each letter into the lock.

STATION 1

Pull 3 N; crate at rest. Friction?

O 3 N E 8 N

STATION 2

Pull 10 N > 8 N limit. Does it slide?

P yes N no

STATION 3

Sliding with 10 N pull, 5 N kinetic, 2 kg. a?

P 2.5 m/s² A 5 m/s²

STATION 4

Sliding, pull 5 N, kinetic 5 N. Motion?

O constant velocity U speeding up

STATION 5

μ = 0.3, normal 20 N. Kinetic friction?

S 6 N T 60 N

STATION 6

3 kg crate: 14 N pull, 5 N kinetic. a?

E 3 m/s² R 4.7 m/s²

LOCK [1] [2] [3] [4] [5] [6]

Hint: what friction does to relative motion.

FRICTION 1 2 3

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Curling stone – the replay

● LIVE

Broadcast analyst desk. A stone slides on ice after release (made-up data).

17 X

SPLIT DATA

t (s)	v (m/s)	a (m/s²)
0	3.0	–
2	2.6	
4	2.2	
6	1.8	
8	1.4	

SLOW-MO v–t GRAPH

STAT 1 · FRICTION'S EFFECT

a = ____ m/s² every interval

STAT 2 · STONE MASS 20 kg

friction = m|a| = ____ N

STAT 3 · STOPS AT

t = ____ s (if a stays the same)

COMMENTARY BOOTH

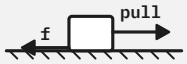
"After release there's no push – only ____ friction, slowing her by ____ m/s every second. That's ____ N on a 20 kg stone, and it should stop near ____ seconds!"

FRICTION 1 2 3

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Picture dictionary: friction

Language goal: I can **tell** static from kinetic friction.



→ ← rub

friction

fricción · atrito

force that opposes slipping

my language: _____

use it: _____



v = 0 · stuck

STUCK

static

estática · estática

not sliding yet; adjusts

my language: _____

use it: _____



SLIDING →

kinetic

cinética · cinética

while sliding

my language: _____

use it: _____



MAX

limit

límite · limite

the most static can give

my language: _____

use it: _____

smooth · small μ

μ × F_N

coefficient μ

coeficiente · coeficiente

grip number, no unit

my language: _____

use it: _____



up close: bumpy

surface

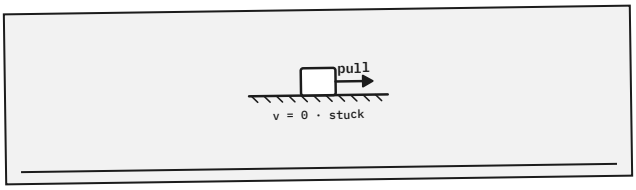
superficie · superfície

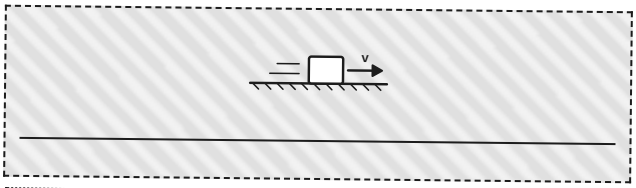
what the object touches

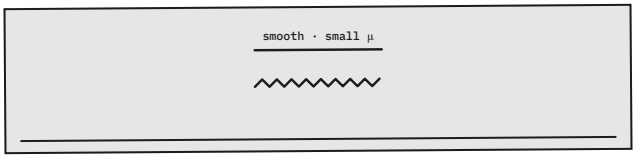
my language: _____

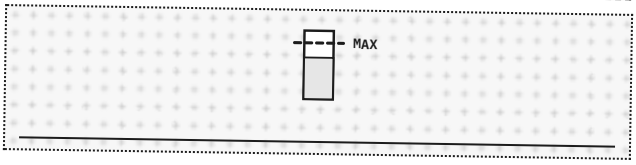
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

FRICTION 1 2 3

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Friction tic-tac-toe

17 D

DIRECTIONS · Complete three in a row. Shade each square you finish.

CALCULATE Pull 4 N, crate at rest, limit 7 N. Actual friction?	**DRAW** Graph friction vs. pull from 0 to 12 N (limit 8 N, kinetic 5 N).	**EXPLAIN** Why doesn't static friction always equal its limit?
REAL WORLD Why do anti-lock brakes avoid skidding?	**CHALLENGE ★** 2 kg crate, μ_k = 0.25, pull 10 N. Find friction (g = 9.8) and a.	**CALCULATE** 3 kg sliding: 14 N pull, 5 N kinetic. Find a.
EXPLAIN Why is it harder to start pushing a couch than to keep it moving?	**DRAW** FBD of a crate at rest with a 3 N pull.	**REAL WORLD** Name a time friction helps you and a time it gets in the way.

Three in a row: _____ · The square that taught me most: _____

FRICTION 1 2 3

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Static or kinetic?

17 N

DIRECTIONS · This crate: static limit 8 N, kinetic friction 5 N. Sort each situation.

A 3 N pull, crate stays put	**B** 8 N pull, crate just barely holds	**C** 10 N pull, crate starts sliding	**D** sliding at constant speed
E crate sitting with no pull	**F** a book resting on a tilted desk	**G** a hockey puck gliding	**H** brakes locked, tires skidding
I shoes gripping as you walk	**J** a box resting on a level floor	**K** a sled sliding downhill	**L** a car parked on a hill

STATIC	KINETIC	NO FRICTION
not sliding yet	already sliding	no tendency to slip

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

FRICTION 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: friction before it slides

17 CS

YOUR READER
A 4th grader who thinks friction only happens once something is moving.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **ORANGE** — A push arrow and a friction arrow in every panel
- ☐ ◌ **YELLOW** — Panel 1's border (the forces are balanced)
- ☐ ◯ **PENCIL** — "static" in panel 1 and "kinetic" in panel 3

1 · PUSH · IT STAYS *Push a heavy box. It does not move.*

CAPTION _____

2 · PUSH HARDER *The box breaks free.*

CAPTION _____

3 · SLIDING *The box slides while you keep pushing.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **static friction · kinetic friction · balanced · push**

_____

_____

FRICTION 1 2 3

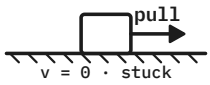
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Doodle Notes · static friction

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Static friction · fricción estática (ES) · atrito estático (PT)



TODAY'S CASE

A box remains at rest as a horizontal push increases from 2 N to 4 N. Both pushes are below the sliding threshold.

01 · SKETCH AND LABEL

Draw an orange push and friction arrow for each trial. Label both arrows in newtons.

02 · CHANGE THE PICTURE

Draw a third moment just after sliding begins. Name the type of friction and explain what changed.

Annotate your sketches. Circle the mark that best shows static friction. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

Before sliding, friction changes to ...

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · static friction

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Static friction · fricción estática (ES) · atrito estático (PT)

A · Static friction is always equal to its maximum.

B · Friction balances a gentle push while the surfaces do not slip.

01 · DEFINITION

What does static friction mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else